



Real-time exam integrity with transparent fairness verdicts

MPOnline Idea & Innovation Hackathon 2026 · Technical Track
PS06 — Resilient & Trustworthy Online Assessment Ecosystem

The Problem

When an exam center drops offline mid-exam, nobody can prove what happened.

- A center loses connectivity for a few minutes. Candidate clocks keep running.
- Candidates don't know if their answers saved. Neither does the control room.
- The only evidence afterwards: a phone log, a coordinator's email, a spreadsheet.
- Because nobody can prove who lost how much time, the safe decision is expensive: **re-conduct the exam for everyone.**

The gap is not detection technology — it's evidence.

The Insight

A disruption is only expensive when it is unprovable.

If, for every candidate, you can show (a) exactly how many seconds they lost, (b) that not one saved answer was lost, and (c) that the record has not been edited since — then the fair remedy is almost never “re-do the exam.” It is “give these three candidates their 47 seconds back.”

So Sentinel is not a monitoring dashboard with an alert. It is an **evidence chain plus a policy engine.**

The Product

One live control tower, five stages, one working vertical slice

Prevention / Detection

Live health across 8 centers & 24 sessions. The server itself notices a missed heartbeat and opens the incident — never the kill switch directly.

Response / Recovery

Affected sessions freeze server-side. Candidates see an honest status message. On reconnect, lost time is restored exactly.

Trust

Every checkpoint is SHA-256 hash-chained, per center. /audit recomputes the whole chain: PASS, or FAIL naming the exact broken row.

Verdict

Re-Conduct / Partial Extension / No Action — computed with the rule, inputs, and arithmetic shown. No black box.

The Live Demo

A judge can break it on stage — and watch it defend itself

1. Control Tower — 8 centers green, 24 live sessions, ledger appending in real time.
2. Press **Kill Switch**. Sentinel detects it, opens a classified incident, freezes sessions.
3. Press **Reconnect**. Sessions resume, time is restored.
4. **Verdict card** appears — Partial Time Extension, with rule, inputs, and arithmetic shown.
5. **/audit** → **Verify** → **PASS**. Then **Tamper** → **Verify** → **FAIL**, naming the exact broken row.

Runs offline, on one laptop, from one command: `npm run build && npm start`

Built & Verified

A running product, not a mockup

- ✓ 14/14 acceptance criteria pass, verified over HTTP/WS and in a real Chromium browser
- ✓ 16 unit tests covering the hash chain, clock, and verdict engine
- ✓ Zero external network calls at runtime — no cloud account, no API keys, no auth to fake
- ✓ One Node.js 22 + TypeScript process: Fastify, raw WebSocket, SQLite, React 19 + Vite
- ✓ Every simulated input is visibly labeled “Simulated telemetry” on screen — we say what is real

Honest Limitations

What we did not build, and why

- **No AI/ML.** The verdict is a deterministic, auditable rule — a decision you can check by hand beats a confidence score you can't. Costs points at an AI-themed event; disclosed on purpose.
- **Illustrative cost figures.** The ■850/candidate re-conduct cost is invented, not sourced from MPOnline — labeled “illustrative” on screen.
- **Unbenchmarked scale.** The architecture doesn't obviously prevent scaling from 8 to 800 centers, but this has not been load-tested.
- **No live deployment.** Blocked by this build sandbox's network policy, not the product — a 6-command fix on an unrestricted network.

What's Next

A realistic one-week plan, not a platform roadmap

- Days 1–2 — Replace the simulator with a real exam-center agent on one real PC
- Day 3 — An integration test for the actual kill→verdict loop
- Day 4 — Per-incident frozen-time accounting (no double-counting on repeat incidents)
- Day 5 — Verdict policy as a signed-off config document, not a code default
- Day 6 — Deploy to a real, shareable URL
- Day 7 — Sit with one real exam control room and validate the verdict card with them

github.com/ramanathanmani/hack · [demo/sentinel-golden-path.webm](#)